SUMMATIVE ASSESSMENT I 2026-27
PHYSICS
CLASS: IX

1. a) Both Statement and Reason are correct, and the Reason explains the Statement

2. b) i-r-y, ii-p-z, iii-q - x

3. c) Distance = 700 m, Displacement = 500 m

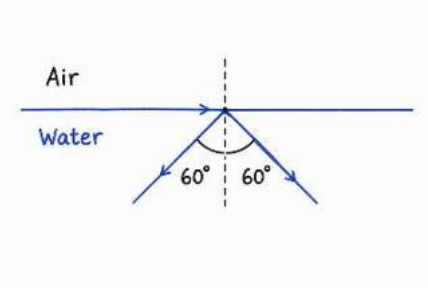
4. b) X-Axis (1 cm = 2s) -Axis (1=10 m)

5. **Uniform velocity:**
   b) A car moving at a constant speed in the same direction along a tarred road.
   d) Light travelling through vacuum.
   Non-uniform velocity:
a) A person walking around a circular stadium at a constant speed.
c) A train coming to a halt at a station.

6. • Cross only at zebra line.
   • Use sidewalks.
   • Look left and right before stepping onto the road.
   • Wear bright clothes.
   • Avoid using the mobile phone. (Any two)

7. While broken lines on the road:
You can safely change lanes, and can cross the line to overtake.
Continuous yellow lines on the road:
Must stay in your lane and not cross or overtake.

8. a) 90°



b)

9. a) Yes, The displacement is the shortest straight-line distance between the initial point and final point. displacement can be zero. The child's initial point and final ending point are the same place. Then the straight-line distance between the initial point and final point is zero.
b) When an object moves in a straight line in only one direction.

10. A.

a) Pour some water into the vessel.
b) It is due to the refraction of light. When a ray of light obliquely enters from one medium to another of different optical densities, it undergoes a deviation at the surface of separation of the medium. Here, the ray of light bends when it passes from water to air

B. It is due to refraction of light as it passes from water into the air. When the light rays reflecting off the fish bend away from the normal when they enter from water to air, it makes the fish appear higher than it really is, closer to the surface.

11. It is due to refraction.
Atmospheric refraction:
• Earth's atmosphere consists of itself act as air medium with different optical densities. When the light ray passes through the atmosphere. It undergoes refraction and bends towards the observer's eyes, making the Sun appear higher in the sky.

12. u = 5 m/s
v = 35 m/s
t = 10 s − 4 s = 6 s

$$a = \frac{v-u}{t}$$
$$= \frac{35-5}{6} = \frac{30}{6} = 5\ m/s^2$$

$$S = ut + \frac{1}{2}at^2$$
$$= 5 \times 6 + \frac{1}{2} \times 5 \times 6^2$$
$$= 30 + 90 = 120\ m$$

Refractive Index:
$$n = \frac{C}{V}$$
$$C = 3 \times 10^8\ m/s$$

Refractive index of water:
$$= \frac{3\times10^8}{2.25\times10^8}$$
$$= 1.33$$

Refractive index of glass:
$$= \frac{3\times10^8}{2\times10^8}$$
$$= 1.5$$

Refractive index of air:
$$= \frac{3\times10^8}{3\times10^8} = 1$$

Ascending Order
Air < Water < Glass

14. a) C

b)

- Reflectors
- Periscope
- Optical fibre cable

c) Total internal reflection

Total internal reflection is not possible always because it requires specific conditions.

1. The ray of light should enter from an optically denser medium to an optically rarer medium.
2. The angle of incidence must be greater than the critical angle.

15

a) Initial time, t1

Second time, t2

Distance covered in the second time = 25 km

Let the initial velocity be .

The speed for the second time .

Distance = Speed × Time

Intial time, $t_1 = 1t$

Second time, $t_2 = \frac{1}{2}t$

Distance covered in the second time = 25 km

Let the intial velocity be u

The speed for the second time $\frac{u}{2}$

Distance = Speed x Time

$$25 = \frac{u}{2} \times \frac{1}{2}$$

$$u = 100\ km/h$$

The distance from A to B

Distance = Speed x Time

=100 x 1

S=100 km

b) It is non-uniform.

16 A.

a) A stone is dropped from a high place.

A car speeding up a height.

When vehicles apply brakes, they slow down.

When vehicles start, they increase their speed.

b) $S = 40\ m$

$t = 7s$

$u = 0\ m/s$

$S = ut + \frac{1}{2}at^2$

$40 = 0 \times 7 + \frac{1}{2} \times a \times 7^2$

$40 = \frac{1}{2} \times a \times 49$

$a = \frac{80}{49} = 1.63\ m/s^2$

B.

a) The rate of decrease in velocity is retardation

OR

Retardation is the negative acceleration

b) $u = 200\frac{m}{s} \quad v = 0 \quad t = 4$

$a = \frac{v-u}{t} = \frac{0-200}{4} = 50$

Retardation = 50 m/s^2

17 A.

a) Refraction

b)

1. Place a glass slab on the letters in a textbook. The letters appear raised.
2. The bottom of a pond appears elevated when viewed from a distance.
3. People engage in bomb fishing aim at a point slightly below the perceived position of the fish.
4. Trembling of stones.

c) It is due to refraction of light. Light rays from the bottom of the pond travel from water to air. As the light ray enters from an optically denser medium to a rarer medium, it bends away from the normal and appears to come from a higher point, and it seems the bottom of the pond is elevated.

B.

a) Refraction

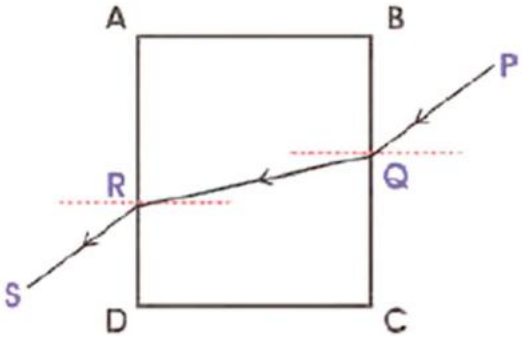
b) No. They work on the principle of reflection

18 A.

a) i = 50°, r = 40°

b) The optical density of medium B is higher than medium A.

c)



B.

a)

1. The light should enter obliquely from one transparent medium to another.
2. The media should have different optical densities.

b)

